

Where to go next

KAUST Academy · Intro to Quantum Computing · your five days end here — the subject doesn't

Keep the memories



quantum.country

Quantum Country — Matuschak & Nielsen

The daily quizzes you did all week were **spaced repetition**: each day re-asked earlier days at growing intervals, because retrieval is what makes memory stick. This site is that same engine — for quantum, forever. An essay-with-built-in-flashcards that emails you when it's time to review. Read one essay, answer honestly, let it schedule you. Twenty minutes a week keeps this course alive.

Read



thomaswong.net

Wong, *Introduction to Classical and Quantum Computing*

Start here after Quantum Country. Free PDF straight from the author's site, written for undergrads, minimal math, runs on Qiskit like our labs. The gentlest real textbook that exists — no library needed.



preskill.
caltech.edu/
ph219

Preskill's Ph219 lecture notes (Caltech)

Free, complete, and beautifully written. The chapter on quantum error correction picks up precisely where today's 1000-physical-to-1-logical story stopped.



michaelnielsen.
org/qcqi

Nielsen & Chuang, *Quantum Computation and Quantum Information*

The textbook — “Mike & Ike.” Everything this week hand-waved, done properly. Chapters 1–2 are exactly our Days 1–2 with the math turned on.

Deeper shelves — your university library has these
Yanofsky & Mannucci, *Quantum Computing for Computer Scientists* — the CS-major's route in. · Zygelman, *A First Introduction to Quantum Computing and Information* — the gentle physics-side twin. · Gerry & Knight, *Introductory Quantum Optics* — where the “machines” morning goes deeper.

Search your campus library portal first: most US universities license Springer Link (Zygelman) and Cambridge Core (Gerry & Knight) — free to you as an enrolled student. Never buy before checking.

Do



learning.
quantum.ibm.com

IBM Quantum Learning + Qiskit tutorials

Structured courses with runnable circuits, on the same free-tier hardware you queued on today. “Basics of quantum information” is the natural next course.



sattamaltwaim.
github.io/
KGSP_Emerging_Technologies_2026

Your molab notebooks

You already own this week's five labs — they stay yours. Re-run them, break them, raise the qubit counts, and watch where the simulator gives out.



qosf.org

QOSF mentorship program

The Quantum Open Source Foundation pairs you with a working researcher for a semester-long open-source project. Free; applications open twice a year.



unitary.fund

Unitary Foundation microgrants

\$4k, no-strings grants for quantum open-source ideas. Undergraduates welcome — several funded projects started as course afterthoughts like this one.



iquhack.mit.edu

iQuHACK & quantum hackathons

MIT's annual quantum hackathon, open worldwide and remote-friendly. A weekend, a team, real hardware credits — the lab, but with prizes.

Join



qiskit.it/join-
slack

Qiskit community Slack

Thousands of practitioners, weekly office hours, and the advocate program once you find your footing. Beginner questions are genuinely welcome in #qiskit-beginners.



arxiv.org/list/
quant-ph/recent

arXiv · quant-ph

Where the field actually publishes, months before journals. Skim weekly; when an abstract makes sense, that's your progress bar moving.

Five days, five ideas

One line each. If a line stops making sense, that's your cue to review — side 1 has the tools.

Day 1 **Entanglement, caught in the act (CHSH)**

Every pre-agreed classical strategy caps at 75% — entangled qubits hit 85%, because the pair is **one shared object**, not two coordinated ones.

Day 2 **Teleportation**

An unknown qubit can't be copied, so it must be **handed over**: Alice's qubit is destroyed, two classical bits fly, Bob's qubit becomes it.

Day 3 **Grover, an interference story**

The oracle flips the marked answer's sign, diffusion reflects about the mean — two mirrors make a rotation, and $\sqrt{N}$ turns land on the answer.

Day 4 **QAOA & the variational loop**

Grade the candidates, then mix; the quantum chip scores knob settings and a classical optimizer steers. Finding the knobs is **the whole game**.

Day 5 **The landscape & the clock**

No hardware winner yet — count **logical** qubits, not headlines — while four independent clocks (algorithms, hardware, deployment, policy) converge on 2029–2030.

“The FORTRAN of quantum hasn't been written yet.”

One of you might write it.